Five remaining PSL_2 extension tables

Private L054 structural recovery component

Theorem 1. *The five current companion tables specified below contain a total of 364 rows. Each structural description has an actual decomposition certificate for its subgroup in the accepted numerical row map, with the explicit Q_{12} convention below. A displayed extension asserts existence of that decomposition; factor names are not asserted to determine a unique group.*

`psigmal227`	38 rows
`psl225d2`	39 rows
`pgammal216`	69 rows
`psigmal225`	103 rows
`pgammal225`	115 rows

Source and acquired inputs. The current tables are attributed to Thomas Connor and Dimitri Leemans. All fifteen native pages were directly inspected, and the first `psl225d2` page was also inspected at higher resolution. We acquire the accepted complete numerical classifications, their exact row bijections, and the named ambient-model transports. We do not assume that all current PDF bytes are identical to a historical 2013 edition. No lattice-property claim follows from the tables' headings.

Every imported proper subgroup and ambient group is given by literal embedded permutations, using either the acquired strict completed-state parser or the literal saved-array parser. There is no reconstruction from a printed group name. The three restriction inputs retain their actual projective defining-model transports. The two full semilinear ambient inventories retain the accepted automorphism-model identification; these are acquired classification interfaces, not conclusions from their orders.

Products associate from the left, as explicitly prescribed in Section 3 of the 2015 naming article. A dot denotes a nonsplit extension, a colon a split extension, and a cross a direct product. Native exponents remain explicit syntax nodes. In `pgammal225`, row 47 is cyclic 26 and row 52 is cyclic 24; neither is an elementary abelian power. In `pgammal216`, row 38 is 2^4. In `psigmal227`, 3^{1+2} is extraspecial of order 27. In `psl225d2`, row 21 is $D_8 \cdot 2$, with a centered dot. For example the top `pgammal216` row means $(\mathrm{L}_2(16) : 2) \cdot 2$. The caption is not substituted for the displayed row.

Reusable literal certificates. Ninety-seven concrete models and their descriptions are acquired from the accepted PSL_2, PGL_2, semilinear, field-extension and A_6-related components. Different concrete models may have the same description. Each use requires an actual isomorphism, checked by the acquired MC.3 criterion: a bijection fixing identity and preserving every right-generator edge is a homomorphism, by induction on words. Factor references point only to acquired or previously verified models. The present application adds 42 models and uses 430 such transports.

The existing split and direct-product criteria apply unchanged. They check a normal subgroup N, a verified complement K, trivial intersection and $|N||K| = |H|$. A colon additionally has a noncommuting pair of factor generators; a cross has all cross-generator pairs commuting. For prime index p, a complement exists exactly when an outside element has order p. The literal reader tests every outside element's pth power. These tests pay 16,368 outside elements in the new models.

Lemma 1 (Composite-quotient lift tests)**.** *Let $N \triangleleft H$ be finite, with $|H : N| = 6$.*

1. *If H/N is cyclic, fix xN generating it. The extension splits if and only if some $a \in Nx$ satisfies $a^6 = 1$.*

2. *Suppose xN, yN generate H/N and satisfy $(xN)^2 = (yN)^3 = (xyN)^2 = N$. The extension splits if and only if some $a \in Nx, b \in Ny$ satisfy $a^2 = b^3 = (ab)^2 = 1$.*

Proof. In the cyclic case, a section supplies a lift of the fixed generator of order six. Conversely any displayed lift with sixth power one has order exactly six, since its image does, and generates a complement.

In the second case the presentation $\langle u, v \mid u^2 = v^3 = (uv)^2 = 1\rangle$ is S_3: $uv = v^{-1}u$ reduces every word to v^i or uv^i, $0 \le i < 3$, and the usual transposition and 3-cycle give all six permutations. Thus the specified quotient is S_3. A complement lifts its two fixed quotient generators and gives the stated relations. Conversely these relations define a homomorphism from S_3 into H. Its image maps onto the quotient of order six, so it has exactly six elements and meets N trivially. Hence it is a complement. □

The literal implementation checks normality first. For the cyclic case it verifies that the six cosets Nx^i cover H and that $x^6 \in N$. For the symmetric case it checks the three quotient relations and that $N, Ny, Ny^2, Nx, Nxy, Nxy^2$ are six disjoint cosets covering H. It then tests all elements of the indicated coset, or all pairs in the two indicated cosets. Each fails at least one required relation. This proves nonsplitting without assuming completeness of a complement-search routine. Eight new composite-quotient models use 39,312 literal lift tests. For instance $(2^4 : 5 : 2) \cdot S_3$ and $D_8 \cdot 6$ require different quotient presentations; the verifier does not identify quotients merely from order six.

The source atom Q_{12}. The three occurrences are `pgammal216`, row 46; `psigmal225`, row 59; and `pgammal225`, row 76. We make the meaning precise as the dicyclic group of order twelve:

$$\langle a, b \mid a^6 = 1,\ b^2 = a^3,\ b^{-1}ab = a^{-1}\rangle. \tag{1}$$

Every word reduces to a^i or a^ib, $0 \le i < 6$, so this presentation has at most twelve elements. The actual certificate gives generators a, b of a twelve-element group with $|a| = 6$ satisfying these relations; therefore it identifies the presented group exactly. The installed GAP primary documentation explicitly defines the dicyclic presentation in this way and reserves generalized quaternion for its 2-power-order case. The source article does not define Q_{12} separately. Our exact presentation fixes the nomenclature qualification rather than claiming that this is a generalized quaternion group of 2-power order.

Finite validation. A bounded witness-generation pass returned all 364 candidates in 10.42 seconds. It uses only the retained groups, not an ambient subgroup census. Normal-subgroup, complement and isomorphism routines propose witnesses; none of their failed searches proves nonsplitting. The independent permutation reader checks complete closures, all row/source joins, 42 new descriptions, 430 isomorphisms and 664,327 right-generator edges. All five top rows have literal certificates, including their nonsplit or split extension assertions.

Independent small controls build the regular dicyclic action directly from multiplication on pairs $(i, j) \in \mathbb{Z}/6 \times \{0, 1\}$:

$$(i, j)(k, l) = (i + (-1)^j k + 3jl \pmod 6,\ j + l \pmod 2).$$

Its central order-two quotient passes the nonsplit S_3 test. The explicit split product $S_3 \times C_2$ fails that test. Likewise C_{12} over its subgroup of order two passes the nonsplit cyclic-six test, while $C_6 \times C_2$ fails. Wrong quotient generators and quotient types are rejected. A dihedral group of order twelve fails the dicyclic relations. Prime-index and direct-product controls are retained as well.

Damaged full certificates alter identity images, right-generator relations, row or element coverage, actual subgroup indices and generators, source powers and dots, product association, acquired model data, factor references, complements, quotient lifts and the dicyclic witness. Every damage must fail its specific check. Ordinary and optimized Python must produce identical mathematical reports. An initial harness expected a later rejection for an invalid S_3 quotient generator; the actual quotient-relation guard rejected it earlier. That harness failure and the corrected expected guard are preserved. No scientific condition was weakened, and the successful discovery was not repeated.

Scope and attribution. This component supplies actual structure witnesses for these 364 current rows, with the explicit Q_{12} presentation qualification, using the previously accepted numerical classifications and literal source maps. The named standard models, original complete inventories and generic transport/decomposition results remain acquired premises. The new lift-test proof, exact applications and controls were authored here under root's registry-reuse task. There is no claim of unique extension labels, new ambient subgroup enumeration, historical byte identity, whole-source completion, question resolution, formalization or measurement credit.